

Krishna Kishore Gedela

Generative AI & ML Engineer

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🔗 Krishna598-DS

🏠 portfolio

Profile Summary

Generative AI Engineer with 2+ years of experience in production-grade backend engineering (Java, Spring Boot, AWS). Expert in architecting end-to-end GenAI systems, including advanced RAG pipelines, multi-agent orchestration via LangGraph, and fine-tuning LLMs with rigorous MLOps observability. M.Sc. in Statistics & Applied Mathematics graduate with strong foundations in probability and linear algebra.

Technical Skills

GenAI & LLMs	LangChain, LangGraph, RAG Systems, OpenAI/Claude API, Fine-tuning, Prompt Engineering
MLOps & Eval	MLflow, LangSmith, RAGAS, Docker, GitHub Actions, CI/CD Pipelines, Render, Railway
Data & ML	FAISS, ChromaDB, PyTorch, Scikit-learn, XGBoost, Hugging Face, Pandas, NumPy, SQL
Backend & Ops	FastAPI, Java, Spring Boot, RESTful APIs, Redis Caching, PostgreSQL, AWS (EC2, S3)
Tools/Other	Git, Streamlit UI, Shell Scripting, Uvicorn, Pydantic, Linux System Administration

Generative AI Projects

- Mar 2026 **Legal Document Intelligence System**, [Live Demo], Python, LangChain, FAISS
- Developed an end-to-end RAG system for legal Q&A using recursive character text splitting and FAISS vector indexing to handle complex legal terminology.
 - Optimized retrieval performance to achieve a **0.942 faithfulness score** and **1.000 context recall** using the RAGAS evaluation framework.
 - Deployed the solution via a FastAPI backend with persistent chat history and auto-generated Swagger docs.
- Apr 2026 **ExamAI: Research System**, [Live Demo], LangGraph, Redis, Docker
- Architected a high-performance 3-agent pipeline (Web Search, RAG, and Verification) using LangGraph to automate complex academic research for JEE/UPSC exams.
 - Leveraged asyncio for parallel agent execution, **reducing response latency by 60%** (25s to 10s) and implemented a self-correction loop for low-confidence outputs.
 - Integrated a Redis caching layer using MD5-hashed query keys, resulting in a **67,000x speedup** for repeated queries.
- May 2026 **LegalMind AI Chatbot**, [Live Demo], GPT-4o-mini, MLflow, LangSmith
- Fine-tuned GPT-4o-mini on a curated dataset of 160+ legal specialized samples, achieving an **85% reduction in training loss** (1.86 → 0.28).
 - Established full MLOps observability using LangSmith for inference tracing and MLflow for experiment tracking.
 - Developed a Streamlit frontend featuring real-time token usage tracking and cost estimation.

Work Experience

- Jun 2024 – Pres. **Software Developer**, Medinovix Pvt. Ltd., Visakhapatnam
- Designed high-concurrency RESTful APIs using Java and Spring Boot; improved PostgreSQL performance by 25% through query optimization and index tuning.
 - Engineered automated system monitoring and DB backup workflows using Shell scripts on AWS EC2, reducing manual server maintenance overhead by 30%.

Education

- 2022 – 2024 **M.Sc. Statistics & Applied Mathematics**, Central University of Tamil Nadu, Thiruvavur
- 2019 – 2022 **B.Sc. Mathematics, Statistics & Computer Science**, Andhra University, Visakhapatnam